

Proposed System

ORCA (Marine EcOsystem Reasoning with Collaborative Agents) is an Agentic AI-powered Marine Decision Intelligence Platform that enables users to interact with Satellite Earth Observation (EO) products, oceanographic observations, weather forecasts, GIS layers, and marine advisories through a natural language interface.

Unlike conventional systems that present independent datasets, ORCA employs a collaborative multi-agent architecture to interpret user intent, retrieve relevant marine information, perform spatial-temporal reasoning, and generate explainable, evidence-based recommendations. By combining Large Language Models (LLMs), Geospatial Intelligence, Explainable AI, and multi-source marine data integration, the platform transforms fragmented marine information into actionable insights for fishermen, researchers, coastal authorities, and maritime operators.

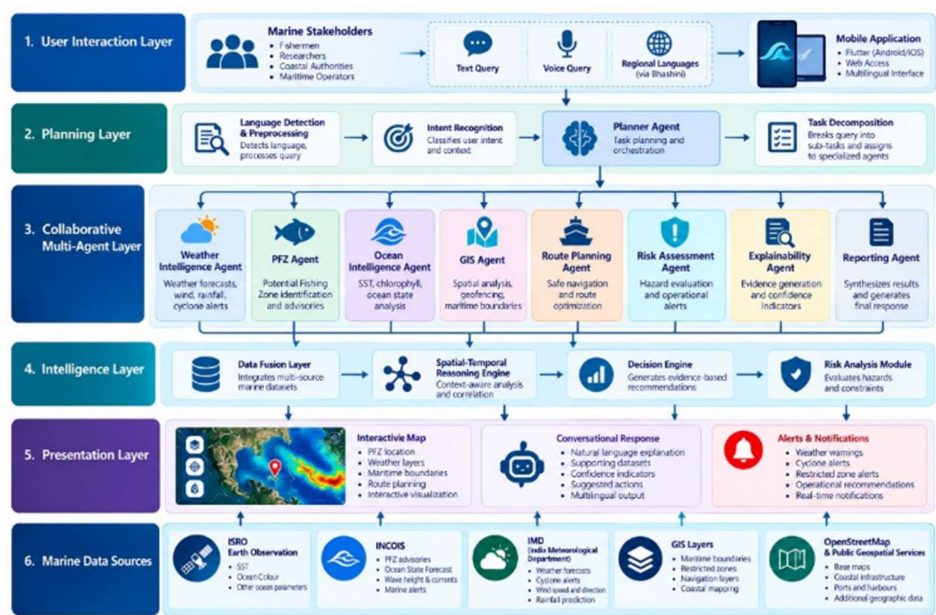


Figure 1.1: Overall ORCA System Architecture

1.2 Overall System Architecture

The ORCA architecture is organized into six functional layers:

Layer	Function

User Interaction	Flutter application, multilingual text & voice input
Planning	Planner Agent, intent recognition, task decomposition
Collaborative Agents	Weather, PFZ, Ocean, GIS, Route Planning, Risk Assessment, Explainability, Reporting
Marine Intelligence	ISRO, INCOIS, IMD, GIS Layers, Maritime Boundaries, OpenStreetMap
Intelligence	Data Fusion, Spatial-Temporal Reasoning, Risk Analysis, Decision Engine
Presentation	Recommendations, Maps, Alerts, Confidence Indicators

1.3 System Workflow

When a user submits a query, ORCA detects the language, identifies the intent, and decomposes the request into domain-specific tasks. Specialized AI agents retrieve relevant information from ISRO, INCOIS, IMD, and GIS sources. The Reasoning Engine performs data fusion and spatial-temporal reasoning, while the Explainability Agent generates supporting evidence. Finally, the Reporting Agent delivers a conversational

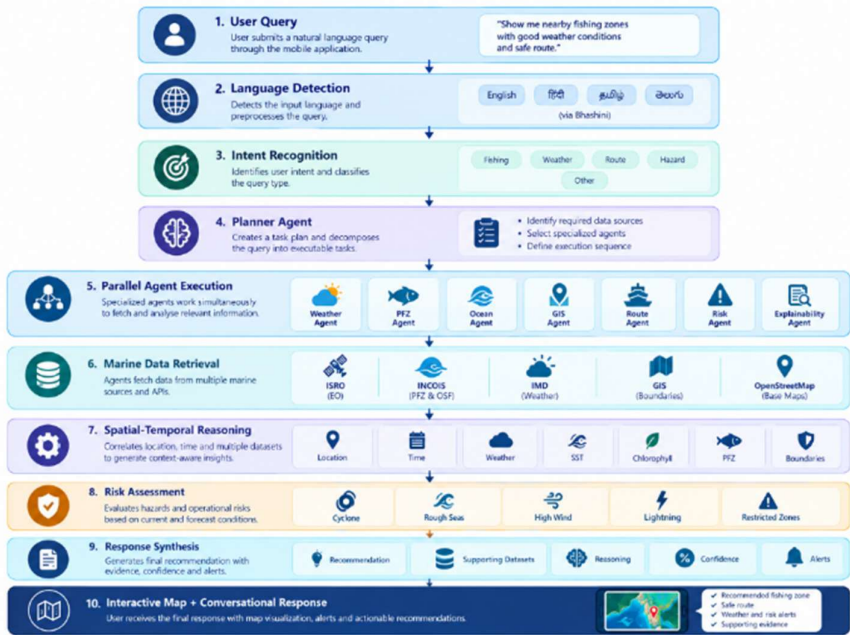


Figure 1.2: ORCA Query Processing Workflow

1.4 Collaborative Multi-Agent Architecture

ORCA follows a collaborative multi-agent architecture, where each agent performs a specialized function while the Planner Agent coordinates workflow execution. This modular design enables parallel processing, scalability, and explainable decision-making.

Table of Collaborative AI Agents

Agent	Primary Responsibility
Planner Agent	Understands user intent, decomposes complex queries, and orchestrates workflow execution.
Weather Intelligence Agent	Retrieves and analyses weather forecasts, wind conditions, rainfall, and cyclone alerts.
PFZ Agent	Identifies Potential Fishing Zones using marine advisories and oceanographic observations.
Ocean Intelligence Agent	Analyses Sea Surface Temperature (SST), chlorophyll concentration, and ocean state parameters.
GIS Agent	Performs spatial analysis, geofencing, and maritime boundary reasoning.
Route Planning Agent	Generates safe navigation routes based on marine conditions.
Risk Assessment Agent	Evaluates hazards such as rough seas, lightning, cyclones, and restricted zones.
Explainability Agent	Produces transparent reasoning, supporting evidence, and confidence indicators.
Reporting Agent	Synthesizes outputs from all agents into a unified conversational response.

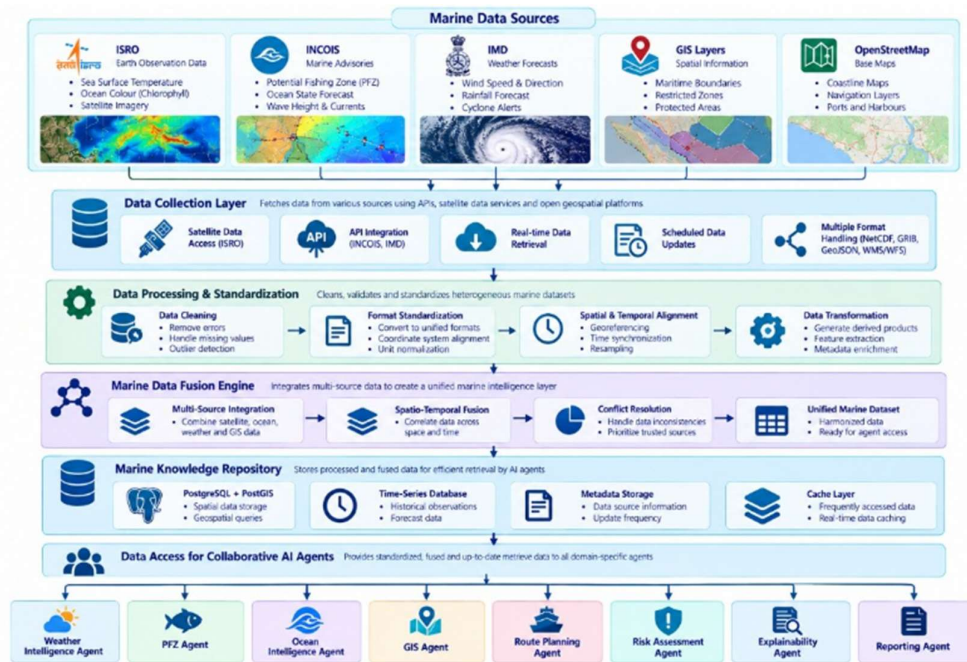
1.5 Marine Data Integration Layer

ORCA integrates heterogeneous datasets from multiple trusted organizations into a unified intelligence layer.

Table 4.2 Marine Data Sources

Source	Information Integrated
ISRO	Satellite Earth Observation (EO), SST, Ocean Colour
INCOIS	Potential Fishing Zone (PFZ), Ocean State Forecast
IMD	Weather Forecasts, Cyclone Alerts, Wind Information
GIS Layers	Maritime Boundaries, Protected Zones
OpenStreetMap	Base Maps and Navigation Layers

A dedicated Data Fusion Layer standardizes and combines these heterogeneous datasets before making them available to the reasoning engine. This enables ORCA to generate recommendations based on correlated observations rather than isolated information.



1.6 Spatial-Temporal Reasoning

The Spatial-Temporal Reasoning Engine combines user location with Potential Fishing Zone (PFZ) advisories, Sea Surface Temperature (SST), chlorophyll concentration, weather forecasts, ocean state, and maritime boundaries to generate context-aware, location-specific recommendations.

1.7 Explainable Decision Support

Every recommendation includes:

- Recommended action
- Supporting datasets
- Reasoning summary
- Confidence indicator
- Marine advisories
- Operational alerts

This Explainable AI approach improves transparency and user trust.

1.8 Multilingual Conversational Interface

ORCA automatically detects the user's language and responds in the same language using multilingual processing, improving accessibility for fishermen and coastal communities.

1.9 Key Features of ORCA

Feature	Capability
Agentic AI	Autonomous planning and task execution
Collaborative AI Agents	Domain-specific reasoning
Natural Language Interface	Text and voice interaction
Multi-Source Data Fusion	EO, weather, oceanographic, and GIS integration
Spatial-Temporal Reasoning	Context-aware recommendations
Explainable AI	Evidence-based decisions
Geospatial Intelligence	Maps and spatial analysis
Geofencing	Boundary and restricted zone alerts
Route Optimization	Safe navigation planning
Multilingual Support	Regional language interaction

Short version:

Backend: Node.js + Express → Render

Cloud Database: Supabase PostgreSQL

Cloud File Storage: Supabase Storage

Authentication: Supabase Auth

AI/Chatflow: Dify

Mobile App: Flutter

So the main flow is:

Flutter → Render Backend → Supabase + Dify + IMD/SACHET/Ocean APIs.